

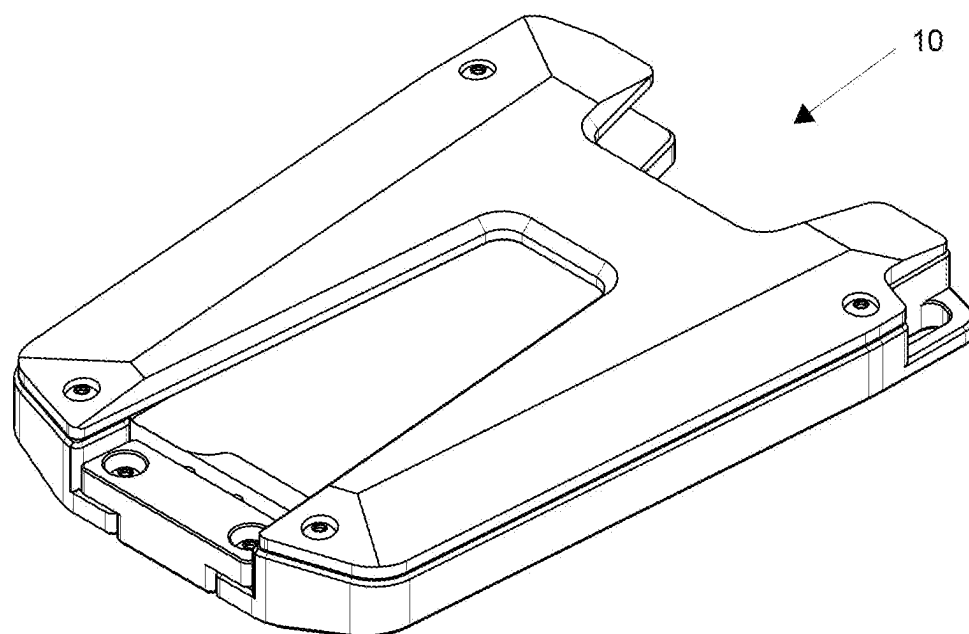


FIG. 1

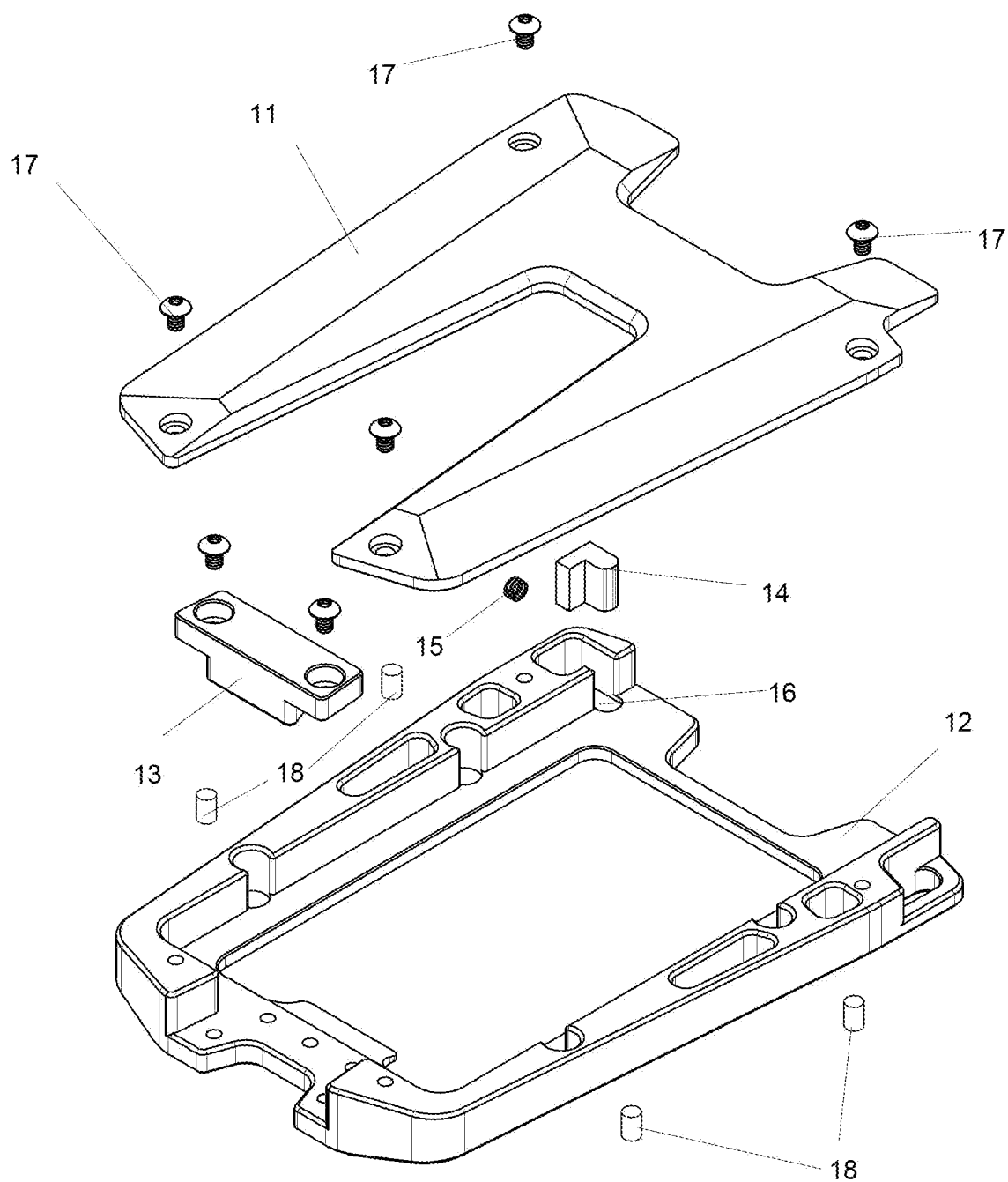


FIG. 2

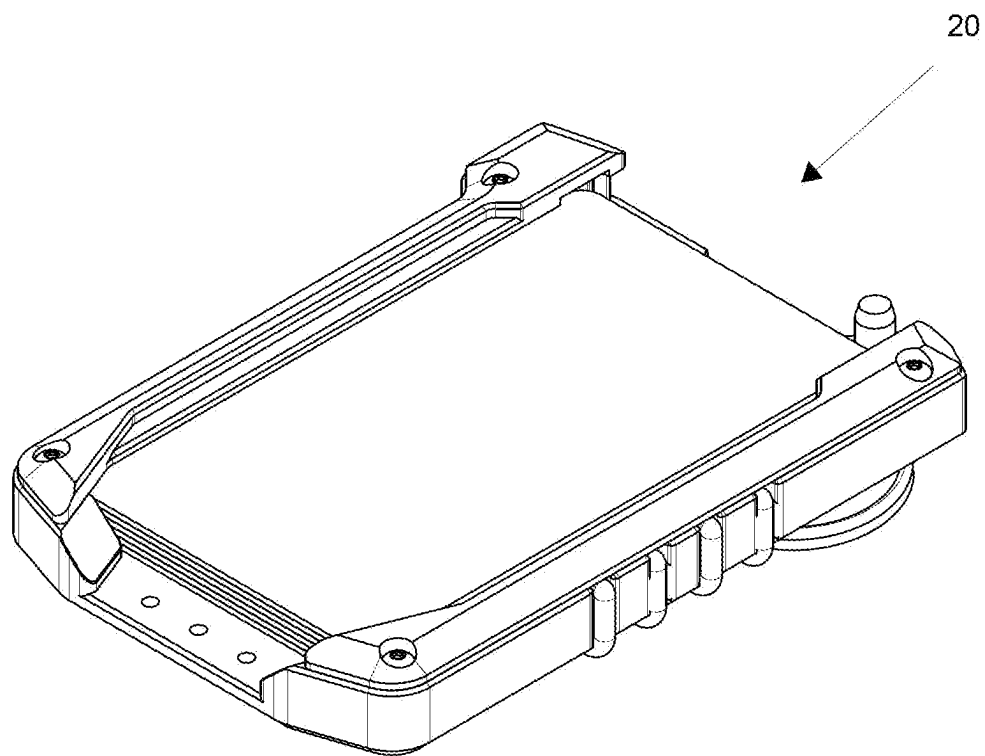


FIG. 3

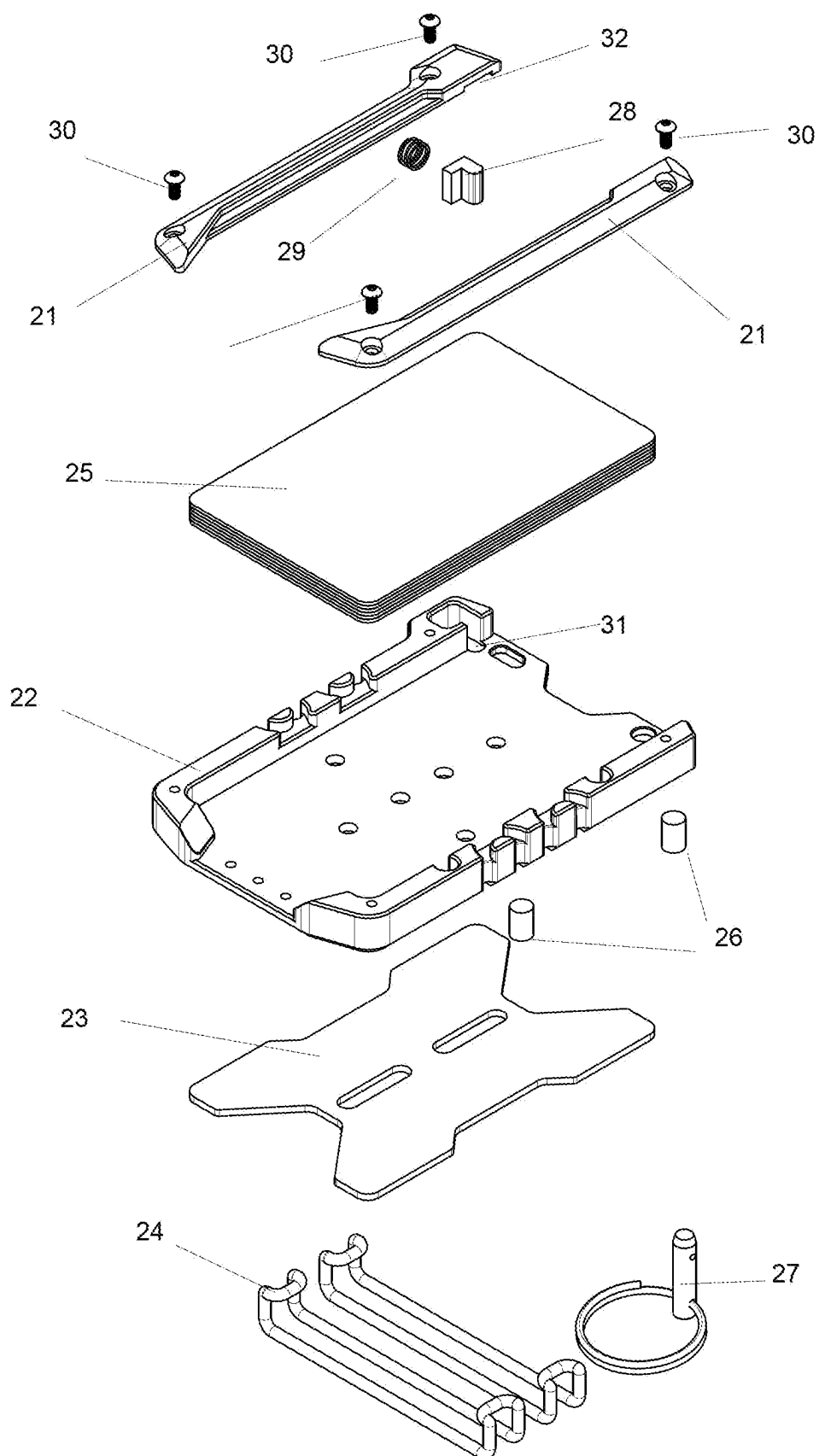


FIG. 4

WALLET WITH CARD HOLDING MECHANISMS

BACKGROUND

[0001] Wallets have been disclosed in the prior art patent literature which are configured for holding cards such as credit cards, identification cards, and the like.

[0002] For example, U.S. Pat. No. 11,653,729 discloses a wallet, comprising: an open-sided shell having a first personal card receiving surface and a second personal card receiving surface facing opposite the first personal card receiving surface, the open-sided shell configured to securably couple at least one personal card along the first personal card receiving surface and the second personal card receiving surface within an internal portion of the open-sided shell, wherein the first personal card receiving surface comprises a first side wall, a second side wall located opposite the first side wall, and a first bottom side wall extending between the first side wall and the second side wall, whereby the first side wall, the second side wall, and the first bottom side wall are configured to retain the at least one personal card in place with respect to the first personal card receiving surface; a first protruding portion coupled to the first side wall and configured to move away from the second side wall to thereby receive the at least one personal card, the first side wall defining a first top portion and a first bottom portion located adjacent the first bottom side wall, the first protruding portion located adjacent the first top portion; and a second protruding portion coupled to the second side wall and configured to move away from the first side wall to thereby receive the at least one personal card, the second side wall defining a second top portion and a second bottom portion located adjacent the bottom side wall, the second protruding portion located adjacent the second top portion.

[0003] However, prior art wallets are often inconvenient to use, and cards are difficult for users to remove from the wallets.

SUMMARY OF INVENTION

[0004] Therefore, an aspect of the present invention is a system of holding cards where the cards are securely held into place but not so secure that they are difficult to remove. Unlike other metal wallets which entirely rely upon stretchable bands to hold cards, the present invention has a front portion that is rigid to allow easy card access. A spring lock achieves a secure hold and easy card access.

[0005] According to an aspect of the present invention disclosed herein, there is provided a wallet with a spring lock card interface, comprising: a surface configured for receiving one or more cards approximately 3.375 inches by 2.125 inches; two side walls on either side of the surface; a top rail; one or more spring locks; one or more springs; one or more buttons connected to the one or more springs; and one or more recesses that allow the one or more buttons to go above and below the one or more cards spaced apart from the top rail.

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. 1 illustrates an embodiment of a wallet according to the present invention.

[0007] FIG. 2 illustrates an exploded view of the wallet of FIG. 1.

[0008] FIG. 3 illustrates another embodiment of a wallet according to the present invention.

[0009] FIG. 4 illustrates an exploded view of the wallet of FIG. 3.

DETAILED DESCRIPTION

[0010] In the present invention, a spring lock card interface portion can be 3d printed from materials such as carbon fiber PLA (pla-cf) or similar 3d printed plastics to reduce card wear over a machined aluminum interface, or using injection molded plastics.

[0011] Spring tension can be varied by inserting a stronger, thicker spring to allow the amount of force required to remove cards to be varied unlike machined aluminum protrusions which offer only a fixed force.

[0012] The spring lock interface of the present invention can be made longer along with sides of wallet to allow 7-20 cards or more on a surface depending on the height the wallet is made. This would be difficult to achieve with machined aluminum as the thin tool required to machine would prevent easy manufacturing the higher they become.

[0013] Silicone bumpers provided in the present invention dampen the sound of card rattle making for a quieter and more pleasing user experience. Silicone bumpers can be removed completely to allow user to remove cards with even less force. The addition of spaces to add up to 4 silicone bumpers is a feature of some embodiments of the present invention.

[0014] Silicone bumpers offer better protection for the edges of credit cards preventing wear on the cards over aluminum protrusions. Both the silicone bumpers and the spring lock can be replaced by the user without purchasing an entire machined part, which can lower replacement part cost.

[0015] The present invention has a hole for a standard Ring-Grip Quick-Release Pin allowing the user to insert a pin to provide extra security from cards becoming dislodged on the front and back due to a sudden impact. It is important to note that this is not needed to hold cards in but is a back up security feature.

[0016] The present invention has O rings that are attached to wallet through internal channels on the side wall and do not affect cards from sliding into the wallet on the front.

[0017] The present invention can have holes (e.g., 6 holes) on the back of the wallet are threaded and spaced to allow existing third party belt, knife and gun holster clips.

[0018] The present invention has a RFID blocking aluminum plate which can be used on the front or back of the wallet. This allows for example a badge to be shown on the front of the wallet while also allowing the belt clip to be used on the back. The present invention has a RFID blocking aluminum plate with slots to allow for the attachment of a police or other badge.

[0019] The present invention has a unique insert tooling system that allows tools of a single function to be able to be used without removing them from the wallet. Tools are screwed down into a receiving area. Functions include glass breaker, 1/4" hex drive, bottle opener inserts.

[0020] FIG. 1 illustrates an embodiment of a wallet according to the present invention.

[0021] Surface 10 is sized for cards that are approximately 3.375 inches wide by 2.125 inches high. This is the same size as a government-issued driver's license. All credit card

companies must comply with these standard dimensions. This ensures all cards are compatible with ATMs and point of sale devices.

[0022] FIG. 2 illustrates an exploded view of the wallet of FIG. 1.

[0023] Reference numeral 11 refers to an upper plate.

[0024] Reference numeral 12 refers to a lower plate.

[0025] Reference numeral 13 refers to an insert tool decorative placeholder where insert tools go.

[0026] Reference numeral 14 refers to a button. Reference numeral 15 refers to a spring. A button and the spring together form a “spring lock”.

[0027] Reference numeral 17 refers to screws which attach the wallet together.

[0028] Reference numeral 16 illustrates one important detail that allows the spring lock to function which are recesses that allows a button to go above and below the cards and not be compressed by the top rail. Without the recesses above and below the tolerances would have to be too tight and the button may not be able to move in and out freely. If the button was made shorter to allow it to move freely without recesses then the cards would catch in the gap between the button and the top and bottom of the wallet and not come out freely. The recesses allow the button to float up and down but not so much as to make a gap where the cards catch.

[0029] Reference numeral 18 refers to silicone bumpers which help prevent cards sliding out and also dampen sound.

[0030] FIG. 3 illustrates another embodiment of a wallet according to the present invention.

[0031] Surface 20 is also sized for cards that are approximately 3.375 inches wide by 2.125 inches high.

[0032] FIG. 4 illustrates an exploded view of the wallet of FIG. 3.

[0033] Reference numeral 21 refers to an upper bar.

[0034] Reference numeral 22 refers to a lower plate.

[0035] Reference numeral 23 refers to an RFID-blocking aluminum back.

[0036] Reference numeral 24 refers to O rings.

[0037] Reference numeral 25 refers to standard-sized cards.

[0038] Reference numeral 26 refers to silicone bumpers which help prevent cards sliding out and also dampen sound.

[0039] Reference numeral 27 refers to a lock pin.

[0040] Reference numeral 28 refers to a button.

[0041] Reference numeral 29 refers to a spring. The spring and button together form a “spring lock”.

[0042] Reference numeral 30 refers to screws which attach the wallet together.

[0043] Reference numeral 31 and 32 illustrate the recess that allows the spring lock to function as described above with reference to FIG. 2.

[0044] The illustrations of embodiments described herein are intended to provide a general understanding of the structure of various embodiments, and they are not intended to serve as a complete description of all the elements and features of apparatus and systems that might make use of the

structures described herein. Many other embodiments will be apparent to those of skill in the art upon reviewing the above description.

[0045] Other embodiments may be utilized and derived from the present invention, such that structural and logical substitutions and changes may be made without departing from the scope of this disclosure.

[0046] Accordingly, the specification and drawings are to be regarded in an illustrative rather than a restrictive sense. Thus, although specific embodiments have been illustrated and described herein, it should be appreciated that any arrangement calculated to achieve the same purpose may be substituted for the specific embodiments shown. This disclosure is intended to cover any and all adaptations or variations of various embodiments.

[0047] Combinations of the above embodiments, and other embodiments not specifically described herein, will be apparent to those of skill in the art upon reviewing the above description. Therefore, it is intended that the disclosure not be limited to the particular embodiment(s) disclosed.

What is claimed is:

1. A wallet with a spring lock card interface, comprising: a surface configured for receiving one or more cards approximately 3.375 inches by 2.125 inches; two side walls on either side of the surface; a top rail; one or more spring locks; one or more springs; one or more buttons connected to the one or more springs; and one or more recesses that allow the one or more buttons to go above and below the one or more cards spaced apart from the top rail.
2. The wallet with a spring lock card interface of claim 1, further comprising one or more bumpers.
3. The wallet with a spring lock card interface of claim 2, wherein the one or more bumpers are silicone.
4. The wallet with a spring lock card interface of claim 2, wherein the one or more bumpers are removable.
5. The wallet with a spring lock card interface of claim 1, further comprising a Ring-Grip Quick-Release Pin.
6. The wallet with a spring lock card interface of claim 1, further comprising one or more O rings attached to the wallet.
7. The wallet with a spring lock card interface of claim 1, further comprising one or more holes on the back of the wallet are threaded and spaced for one or more of a belt, knife and gun holster clip.
8. The wallet with a spring lock card interface of claim 1, further comprising an RFID blocking plate on the front or back of the wallet.
9. The wallet with a spring lock card interface of claim 1, further comprising insert tooling system that allows one or more tools to be able to be used without removing them from the wallet.
10. The wallet with a spring lock card interface of claim 9, wherein the one or more tools are screwed down into a receiving area.

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